Question 1 [10 Points] [LO's: 2, 4]

a- For LTIS define the following items: Stability, ROC and Difference Equation.

b- For the following transfer function:

$$H(z) = \frac{z^2}{(z - 0.5)(z - 1)^2}$$

(i) Plot the pole - zero diagram of H (z).

(ii) Find the difference equation of this filter, y (n).

(iii) Represent the filtering operation in signal flow diagram form.

c- Using residue method find "4" values for impulse response, h(n), of this system.

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for the following : $H(Z) = \frac{Z^2}{(Z-0.5)(Z-1)^2}$

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(a) Plot the Pole_Zero diagram
(b) Find the difference equation of this filter, $y(n)$
(c) Represent the filtering operation in Signal flow diagram
(d) using residue method Find "4" values for Impulse response $h(n)$ of this system.

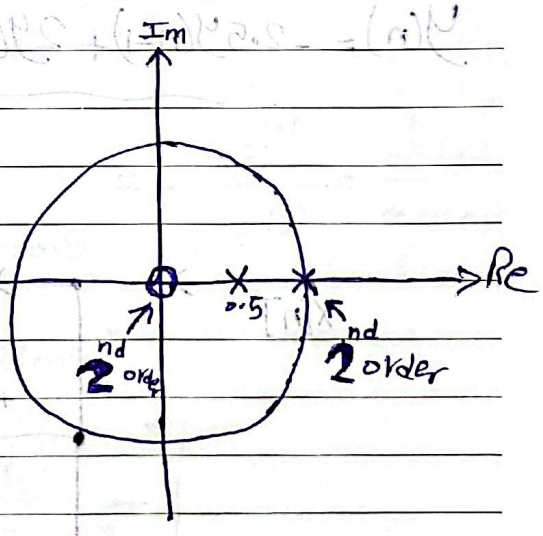
Solution

$$H(Z) = \frac{Z^2}{(Z-0.5)(Z-1)^2}$$

Zeroes exists at $Z = \{0, 0\}$
Poles exists at $Z = \{0.5, 1, 1\}$

System Stability :

- Critically stable



difference equation

$$H(Z) = \frac{Z^2}{(Z-0.5)(Z-1)^2} = \frac{Y(Z)}{X(Z)}$$

$$= \frac{Z^2}{(Z-0.5)(Z^2-2Z+1)} = \frac{Z^2}{Z^3-2Z^2+Z-0.5Z^2+Z-0.5}$$

difference equation طلع معنا بسط z^{-3} نفرض في

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$$\therefore H(z) = \frac{z^2}{z^3 - 2.5z^2 + 2z - 0.5}$$

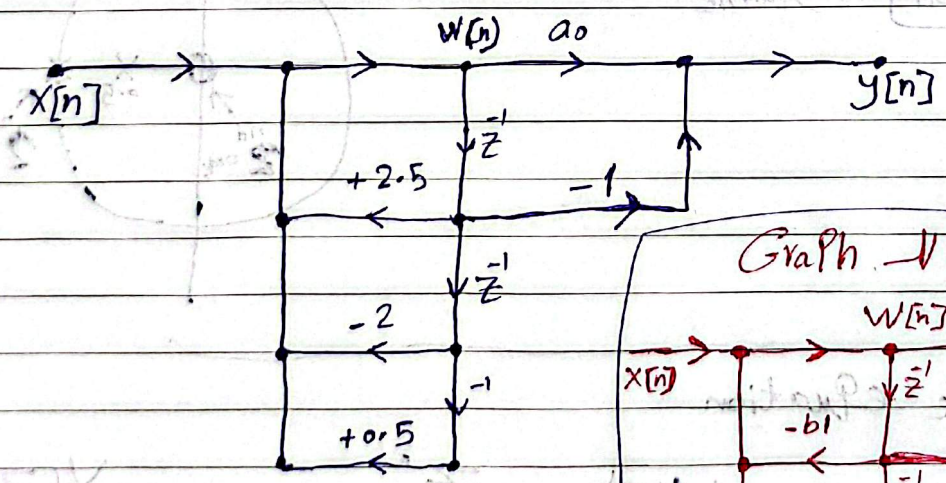
$$\therefore H(z) = \frac{z^{-1}}{1 - 2.5z^{-1} + 2z^{-2} - 0.5z^{-3}} = \frac{Y(z)}{X(z)}$$

$$\Longrightarrow z^{-1}X(z) = Y(z) - 2.5Y(z)z^{-1} + 2Y(z)z^{-2} - 0.5Y(z)z^{-3}$$

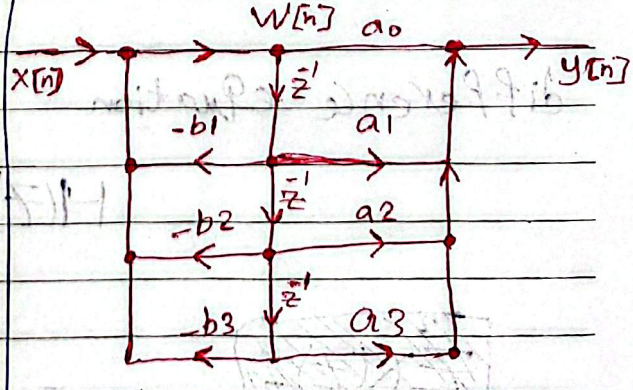
خاصية $\Rightarrow m(n-k) \Longleftrightarrow M(z)z^{-k}$

$$X(n-1) = y(n) - 2.5y(n-1) + 2y(n-2) - 0.5y(n-3)$$

$$y(n) = -2.5y(n-1) + 2y(n-2) - 0.5y(n-3) - x(n-1)$$



Graph الشكل المطلوب



$$\frac{a_0 + a_1z^{-1} + a_2z^{-2} + a_3z^{-3} + \cdots}{1 + b_1z^{-1} + b_2z^{-2} + b_3z^{-3} + \cdots}$$

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Using Residue Method find "4" Values for Impulse Response h(n).

$$H(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

$$\Rightarrow \text{Solve: } F(z) = z^{n-1} H(z)$$

$$\therefore F(z) = \frac{z^{n-1} z^2}{(z-0.5)(z-1)^2} = \frac{z^{n+1}}{(z-0.5)(z-1)^2}$$

* F(z) has Pole at $z=0.5$ and 2nd order Pole at $z=1$

$$\therefore h(n) = Res[F(z), P_1] + Res[F(z), P_2]$$

$$\therefore Res[F(z), 0.5] = \left.\frac{(z-0.5)\, z^{n+1}}{(z-0.5)(z-1)^2}\right|_{z=0.5} = \frac{0.5^{n+1}}{(0.5-1)^2} = \frac{0.5(0.5)^n}{+0.25}$$

$$\Rightarrow Res[F(z), 0.5] = \frac{0.5}{0.25} * 0.5^n = 2(0.5)^n$$

$$Res[F(z), 1] = \frac{d}{dz}\left[\frac{z^{n+1}(z-1)^2}{(z-0.5)(z-1)^2}\right]$$

$$= \left.\frac{(z-0.5)(n+1) * z^n - z^{n+1}(1)}{(z-0.5)^2}\right|_{z=1}$$

$$= \frac{0.5(n+1) - (1)}{0.25} = \frac{0.5n + 0.5 - 1}{0.25}$$

$$= 4(0.5n - 0.5) \quad 2n - 2 = 2(n-1)$$

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$\therefore h(n) = 2(n-1) + 2(0.5)^n$

first 4.. Values $n = 0, 1, 2, 3$

$h(0) = 2(0-1) + 2(0.5)^0 = -2 + 2 = 0$

$h(1) = 2(1-1) + 2(0.5)^1 = 0 + 1 = 1$

$h(2) = 2(2-1) + 2(0.5)^2 = 2 + 0.5 = 2.5$

$h(3) = 2(3-1) + 2(0.5)^3 = 4 + 0.25 = 4.25$